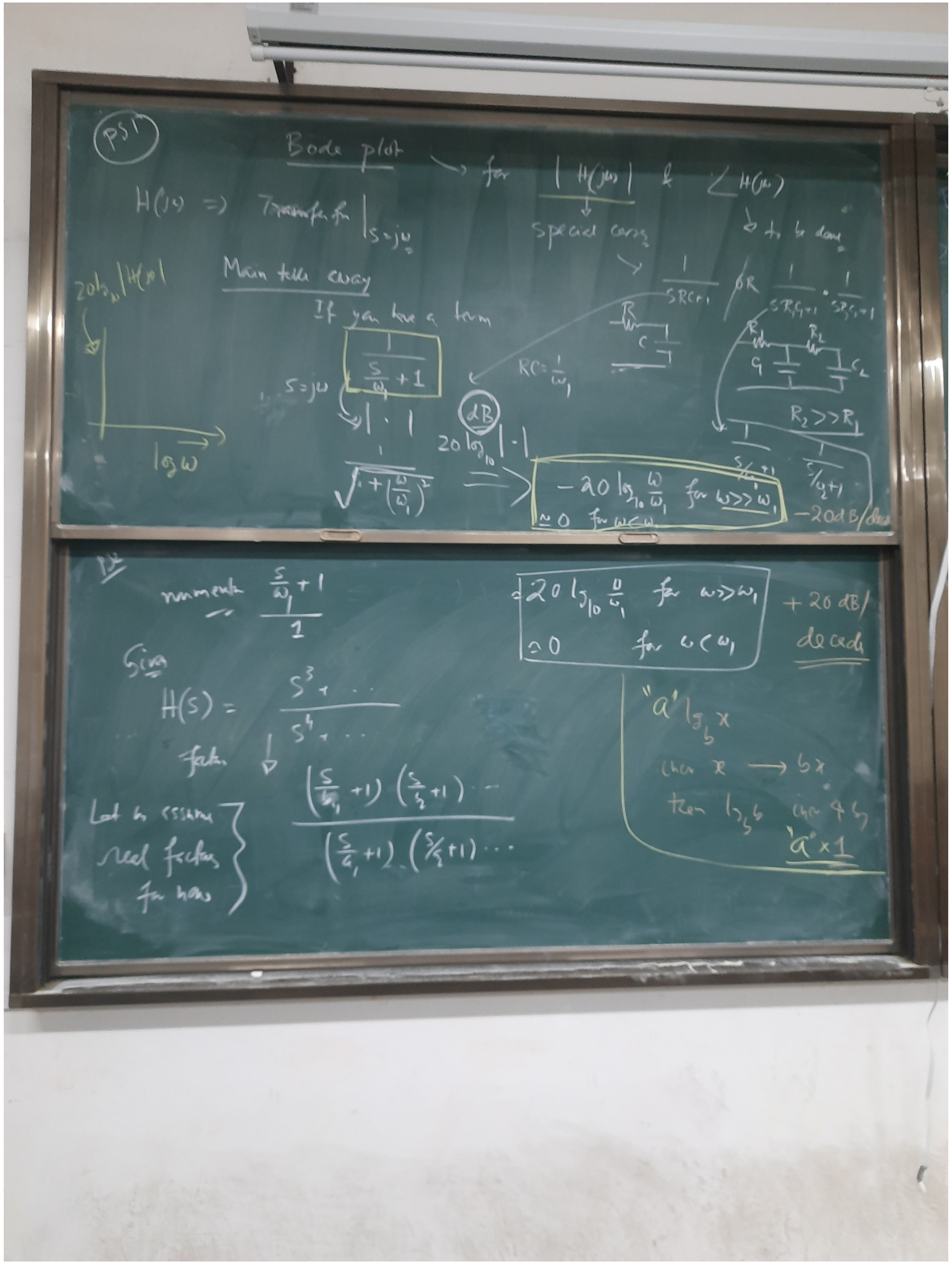


PS1

Bode plot → for $|H(j\omega)|$ & $\angle H(j\omega)$

$|H(j\omega)|$ ↓ special cases → $\frac{1}{sRC+1}$ OR $\frac{1}{sR_1C_1+1} \cdot \frac{1}{sR_2C_2+1}$

$\angle H(j\omega)$ ↓ to be done

$H(j\omega) \Rightarrow$ Transfer fn $\big|_{s=j\omega}$

Main take away

If you have a term $\boxed{\frac{1}{\frac{s}{\omega_1}+1}}$

$s=j\omega$ → $|\cdot| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_1}\right)^2}}$

dB

$20\log_{10}|\cdot| \Longrightarrow \boxed{\begin{array}{l} -20\log_{10}\frac{\omega}{\omega_1} \text{ for } \omega >> \omega_1 \\ \simeq 0 \text{ for } \omega << \omega_1 \end{array}}$ -20 dB/dec

$RC = \frac{1}{\omega_1}$

$R_2 >> R_1$

$\frac{1}{s/\omega_1+1} \cdot \frac{1}{s/\omega_2+1}$

PS2

numerator $\frac{\frac{s}{\omega_1}+1}{1}$

$\boxed{\begin{array}{ll} 20\log_{10}\frac{\omega}{\omega_1} & \text{for } \omega >> \omega_1 \\ \simeq 0 & \text{for } \omega < \omega_1 \end{array}}$ $+20$ dB/decade

Given

$$H(s) = \frac{s^3+\cdots}{s^4+\cdots}$$

factor ↓

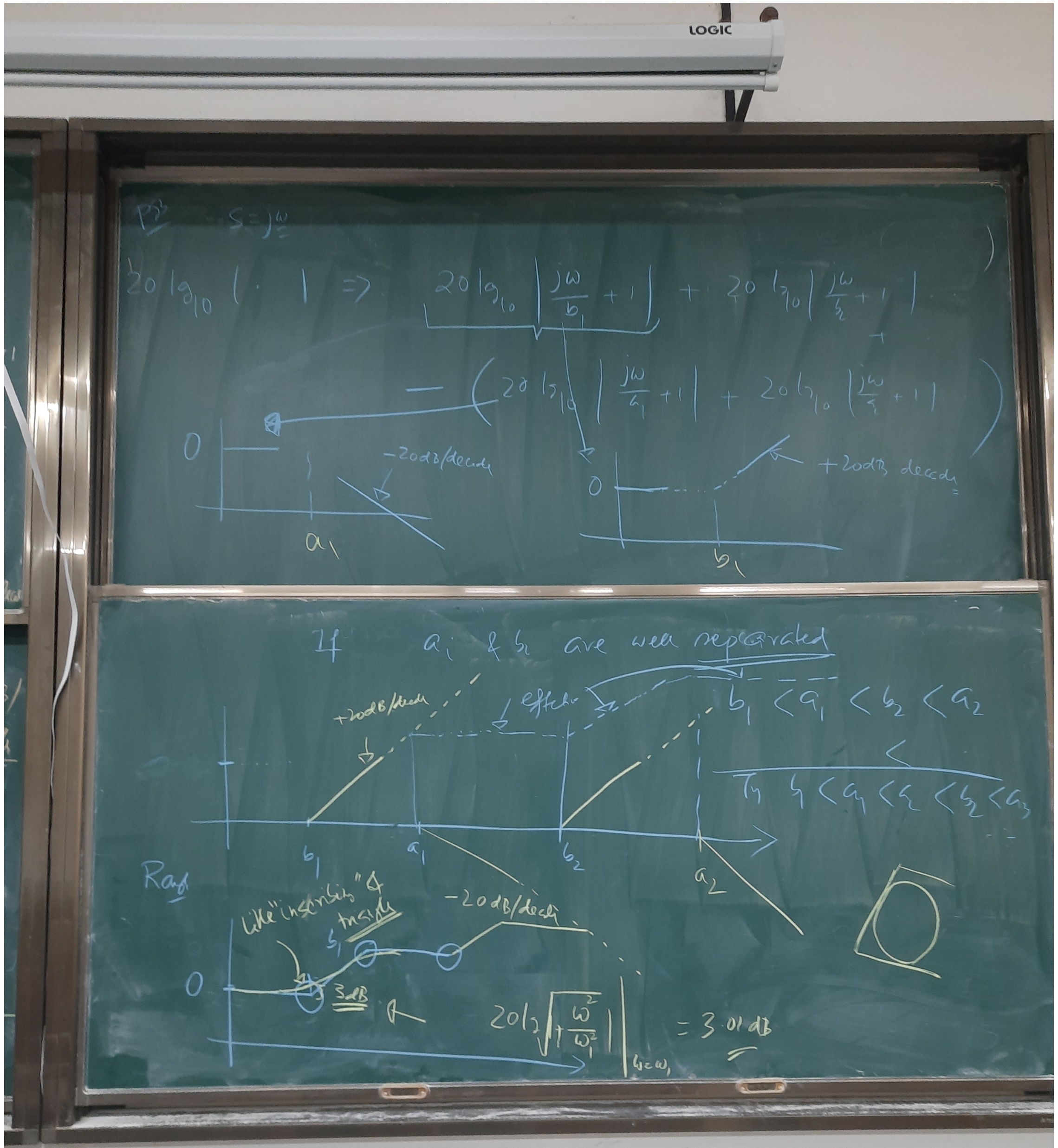
$$\frac{\left(\frac{s}{\omega_1}+1\right)\left(\frac{s}{\omega_2}+1\right)\cdots}{\left(\frac{s}{a_1}+1\right)\left(\frac{s}{a_2}+1\right)\cdots}$$

Let us assume real factors for now

"a" $\log_b x$

change $x \longrightarrow bx$

then $\log_b b$ change by "a" × 1



Pg: $s = j\omega$

$$20\log_{10}|\cdot| \Rightarrow \underbrace{20\log_{10}\left|\frac{j\omega}{b_1}+1\right|} + 20\log_{10}\left|\frac{j\omega}{b_2}+1\right| \cdots$$

$$-\left(20\log_{10}\left|\frac{j\omega}{a_1}+1\right| + 20\log_{10}\left|\frac{j\omega}{a_2}+1\right|\right)$$

0
a_1
−20dB/decade
0
b_1
+20dB decade

If a_i & b_i are well separated

$b_1 < a_1 < b_2 < a_2$

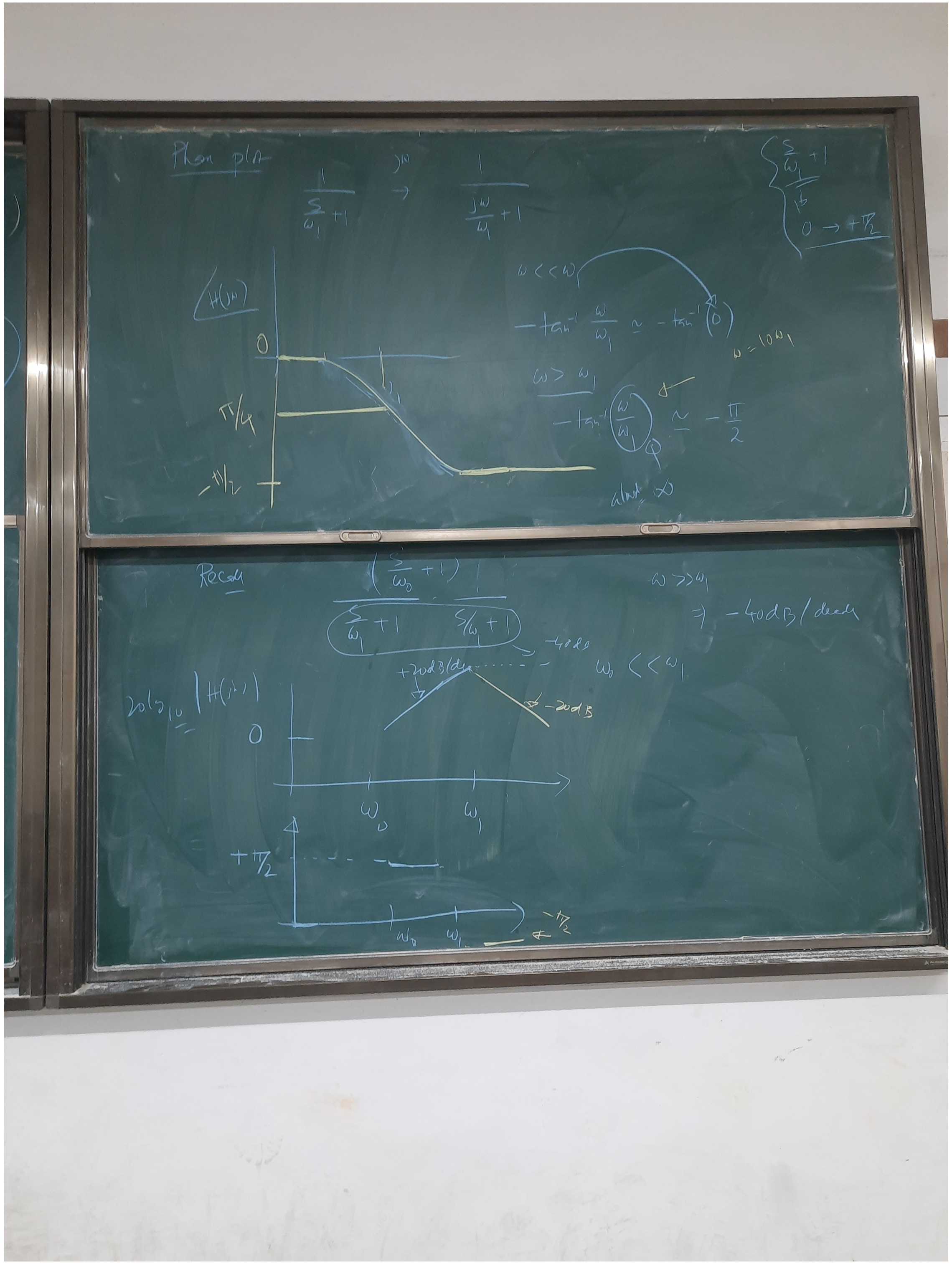
Ty $b_1 < a_1 < a_2 < b_2 < a_3$

+20dB/decade
effect
b_1
a_1
b_2
a_2

Real

like "inscribing" & inside
−20dB/decade
b_1
0
3dB

$$20\log_{10}\sqrt{1+\frac{\omega^2}{\omega_1^2}}\,\Bigg|_{\omega=\omega_1} = 3.01\text{ dB}$$



Phase plot

$$\frac{1}{\frac{s}{\omega_1}+1} \xrightarrow{j\omega} \frac{1}{\frac{j\omega}{\omega_1}+1}$$

$\left\{ \frac{s}{\omega_1}+1 \right.$

$0 \to +\pi/2$

$\angle H(j\omega)$

$\omega \ll \omega_1$

$$-\tan^{-1}\frac{\omega}{\omega_1} \simeq -\tan^{-1}(0)$$

$\omega = 10\omega_1$

$\omega > \omega_1$

$$-\tan^{-1}\left(\frac{\omega}{\omega_1}\right) \simeq -\frac{\pi}{2}$$

almost ∞

0
$\pi/4$
$-\pi/2$
ω_1

Recall

$$\frac{\left(\frac{s}{\omega_0}+1\right)}{\boxed{\frac{s}{\omega_1}+1}} \cdot \frac{1}{\boxed{s/\omega_1+1}}$$

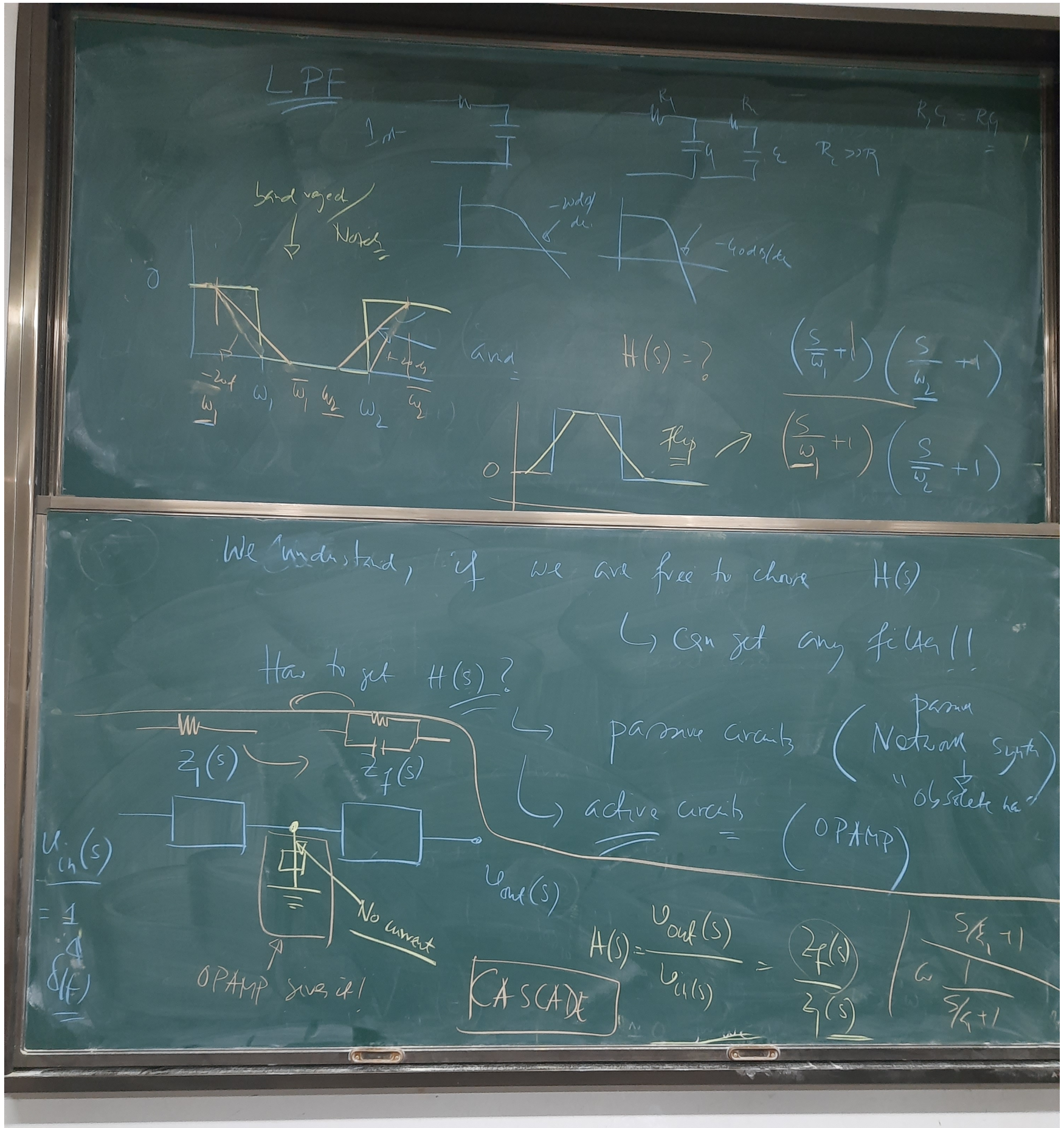
$\omega \gg \omega_1$

$\Rightarrow -40dB/decade$

$\omega_0 \ll \omega_1$

$20\log_{10}|H(j\omega)| =$

-40dB
+20dB/dec
-20dB
0
ω_0
ω_1
$+\pi/2$
$-\pi/2$
ω_0
ω_1


LPF
1st
R_1
R_2
C_1
C_2
$R_2 >> R$
$R_1C_1 = R_2C_2$
-20dB/dec
-40dB/dec
band rejected
Notch
0
-20dB
-40dB
ω_1
$\bar{\omega}_1$
$\underline{\omega_2}$
ω_2
$\bar{\omega}_2$
and
Flip
0
$Z_1(s)$
$Z_f(s)$
$v_{in}(s)$
$v_{out}(s)$
No current
OPAMP saves it!
$s/\omega_1 + 1$
ω
$\frac{1}{s/\omega_1+1}$

$H(s) = ?$

$$\frac{\left(\frac{s}{\bar{\omega}_1}+1\right)\left(\frac{s}{\underline{\omega_2}}+1\right)}{\left(\frac{s}{\underline{\omega_1}}+1\right)\left(\frac{s}{\omega_2}+1\right)}$$

We understand, if we are free to choose H(s)

↳ Can get any filter!!

How to get H(s)?

↳ passive circuits (passive Network synth) ↓ "obsolete now"

↳ active circuits = (OPAMP)

$v_{in}(s) = 1 \Updownarrow \delta(t)$

$$H(s) = \frac{v_{out}(s)}{v_{in}(s)} = \frac{Z_f(s)}{Z_1(s)}$$

CASCADE